GRADE 7 LIFE SCIENCE

Trait Inheritance

Genes, chromosomes, and evidence

Learning goal

Use scientific evidence to explain how genes and chromosomes contribute to the inheritance of a specific trait.

Mock lecture notes prepared for the ExitLoop teacher demo

CORE IDEAS

What carries inherited information?

- A gene is a section of inherited information related to a trait. The gene is not the visible trait itself.
- Genes are located at specific positions on chromosomes.
- An offspring receives one chromosome of a pair from each parent, so both parents contribute gene versions.
- A scientific explanation connects evidence about the inherited gene versions to the trait that develops.

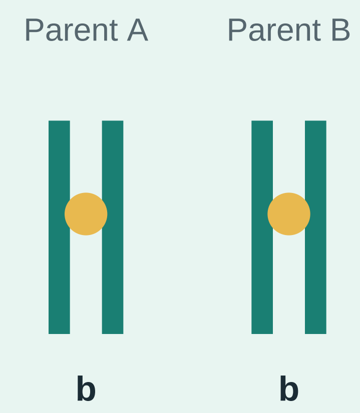
CLASS EXAMPLE

A fictional beetle trait

Evidence from the case

- The bristle-shape gene is located on chromosome 3.
- Each parent has the gene versions B and b.
- The offspring inherited one chromosome 3 from each parent and has b and b.
- Beetles with b and b develop curved bristles. Beetles with at least one B develop straight bristles.

Trace the chromosomes



b + b

curved bristles

EXPLANATION TASK

Use evidence and reasoning

Student question

Explain how the offspring inherited curved bristles. Describe the roles of the gene and chromosomes, and use at least two facts from the beetle case as evidence.

A strong explanation connects

- gene information to the trait
- the gene to a chromosome
- one contribution from each parent
- case facts to the conclusion

Related new example

A seed-coat gene is on chromosome 2. A seed inherits c from each parent, and plants with c and c develop spotted coats. Explain how the seed inherited the spotted-coat trait using the same relationships.

TEACHER PLANNING

Ideas to check in student explanations

Possible missing relationships

- The response names the trait but does not explain what information the gene carries.
- The response mentions genes and chromosomes but never connects their relationship.
- The response follows only one parent's contribution.
- The response repeats facts without explaining how they support the conclusion.

Possible incorrect ideas

- The gene is the visible trait itself.
- Genes and chromosomes are unrelated inheritance systems.
- Only one parent determines this trait.
- A physical change acquired by a parent changes the gene versions passed to offspring.

Content review references: Georgia CASE standards portal; AAAS Project 2061 assessment topics; Genetics Concept Assessment design literature.